

Winning Back the Base

A Value-Based Customer Retention Strategy for Company A

Turning 100,000 customer records into a quantified, ROI-positive retention program — targeting not just who will churn, but who is most valuable to keep.

0.693

XGBoost ROC-AUC (CV-validated)

\$9.8M

net benefit / yr (per 1M subs)

1.6x

more revenue saved vs. churn-only

Executive summary

Company A loses customers faster than it can profitably replace them. We can fix that.

■ The problem

In a saturated market, ~95% of growth comes from switchers and re-acquiring a lost customer costs 5–25x more than keeping one. Churn is Company A's most expensive leak.

■ What the data says

Churn is predictable and concentrated. The #1 driver is aging handsets (>12 months old churn at ~57–59% in our sample) — a lever Company A directly controls.

■ Our solution

A monthly value-based targeting engine: rank customers by Expected Value-at-Risk (churn probability x customer value), then make proactive, persona-specific retention offers.

What it delivers — per 1,000,000 subscribers

Best model	XGBoost · AUC 0.693
Targeting edge	captures 45% of revenue-at-risk in the top 20%
Net benefit / yr	\$9.8M (1.64x ROI)
Customers saved / yr	~15,490
Our fee	\$840k/yr · 8.5% of value created

Sources: Bain & Company; Harvard Business Review; analysis of Company A dataset (n=100,000).

Market analysis: retention is the only profitable growth lever

\$344B

US wireless market (2025)

1.0%

Best-in-class monthly churn

5–25x

Re-acquire vs. retain cost

25%+

Profit lift from +5% retention

■ **The market is saturated**

Penetration exceeds 100%. Carriers grow almost entirely by stealing switchers — making every avoided cancellation worth far more than a new-customer ad.

■ **Acquisition is expensive**

Industry CAC runs into the hundreds of dollars per line; retention offers are a fraction of that, so saved customers are immediately accretive.

■ **Small churn moves, big profit**

Bain's classic finding: a 5-point retention improvement can lift profits 25–95%. Churn is the highest-leverage metric in the P&L.

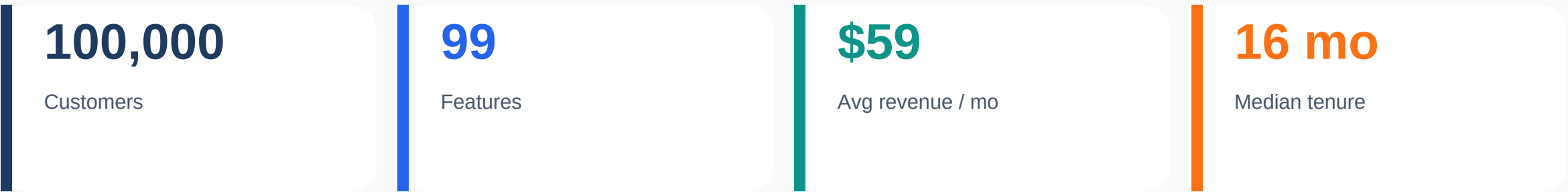
■ **Loyalty compounds**

Tenured customers spend more, cost less to serve, and refer others — so protecting the base protects future revenue too.

Sources: industry market sizing 2025; Verizon/T-Mobile investor reports; Bain & Company; Reichheld, HBR.

The client & the data

Company A is a US wireless carrier facing elevated churn. We received two linked tables covering 100,000 customers — behavioural usage and customer profile — joined on Customer_ID.



Behavioural (Record.csv)

Minutes of use, overage, revenue, call quality, recharges, and the churn label — the in-life signals of disengagement.

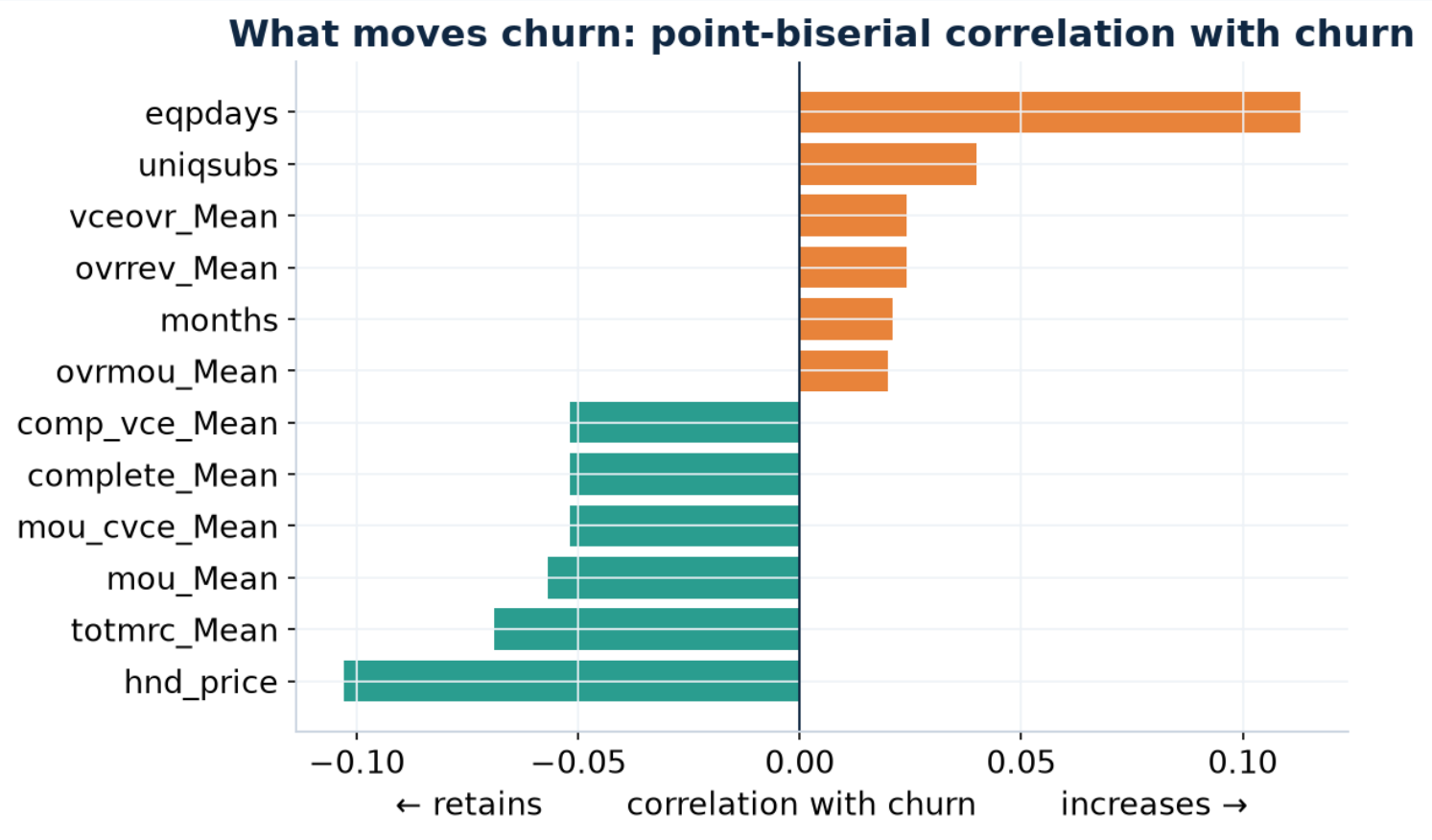
Profile (Client.csv)

Handset age & price, refurbished flag, web-capability, credit, tenure and demographics — the structural drivers.

Data-quality note: the modelling sample is balanced ~50/50 by design (oversampled), so it is used for ranking — not as the real churn rate. Demographic fields are 25–49% missing and are de-emphasised in favour of controllable, behavioural drivers.

Source: Company A dataset (Client.csv + Record.csv), 100,000 customers.

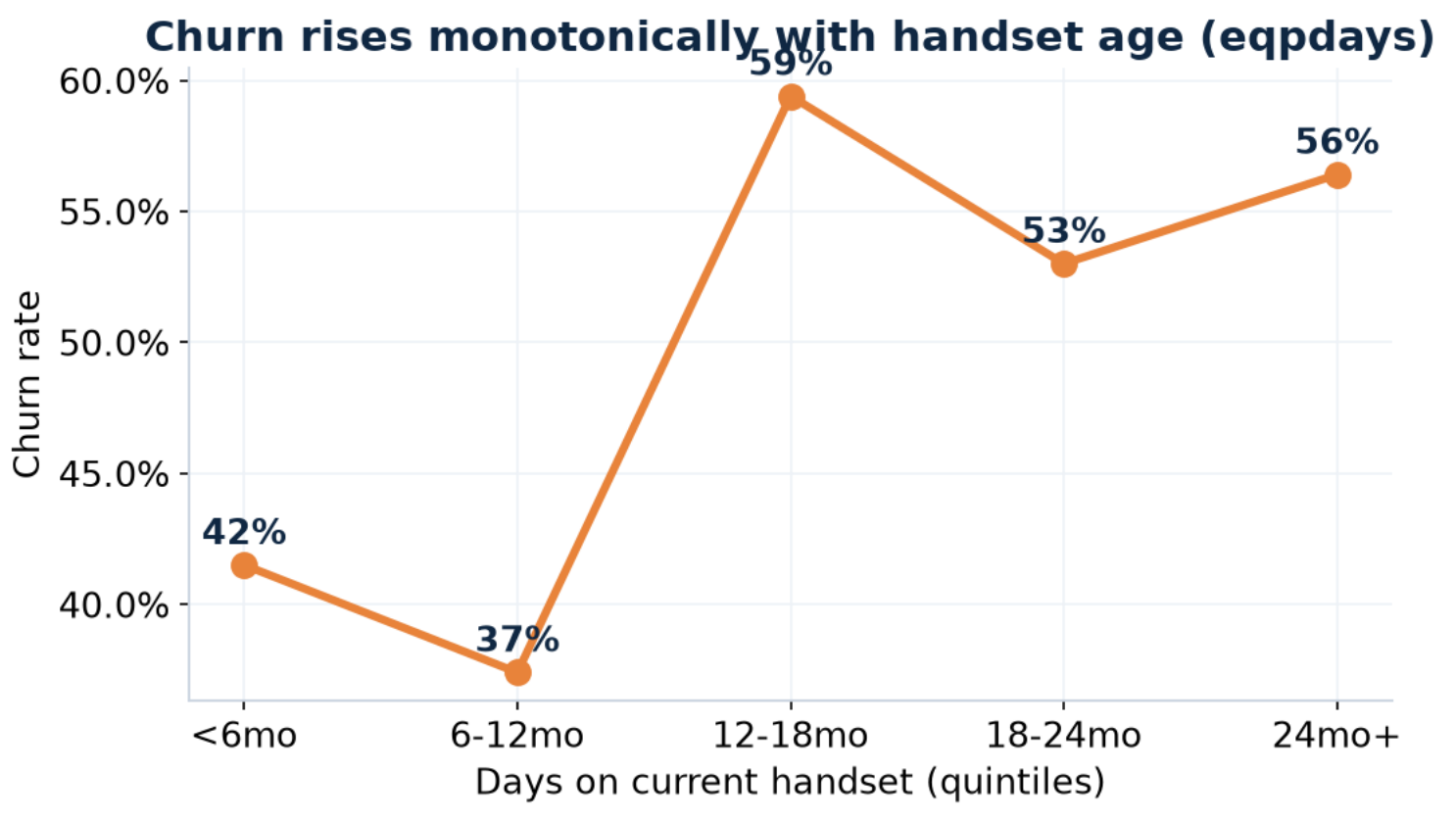
What the data reveals: who leaves, and why



- Equipment age is the #1 signal**
Longer time on the same handset (eqpdays) is the strongest positive correlate of churn (+0.11) — phones age out, customers shop around.
- Investment buys loyalty**
Higher handset price and higher recurring charge both correlate negatively with churn — customers with newer, pricier phones stay.
- Engagement matters**
More minutes of use and completed calls track with lower churn; declining usage is an early warning sign.

Source: Pearson correlations with churn, Company A dataset (n=100,000).

The headline insight: the handset-age cliff



- A clear inflection at ~12 months**
Churn jumps from ~37% (6–12 mo) to ~59% once a handset passes 12 months, and stays elevated thereafter.
- This is an actionable lever**
Unlike age or income, handset age is something Company A can change — through proactive upgrade and trade-in offers.
- Refurbished handsets churn more**
Refurb lines churn at ~53% vs ~49% for new — a quality-of-experience flag worth fixing.

Source: churn rate by handset-age bucket and refurbished flag, Company A dataset.

Framing the business problem as an ML task

Business question: “Which customers are about to leave, how much revenue do they put at risk, and what is the most cost-effective way to keep them?”

- **ML formulation**

A supervised binary classification problem: predict $P(\text{churn})$ per customer from behavioural + profile features.

- **But probability isn't enough**

A \$130/mo customer and a \$25/mo customer are not equally worth saving. We rank by Expected Value-at-Risk = $P(\text{churn}) \times \text{customer value}$.

- **Primary metric: ROC-AUC**

Ranking quality matters more than a single cutoff, because we act on the highest-risk, highest-value customers first.

- **Success = profit, not accuracy**

The model is judged by incremental margin and avoided re-acquisition cost from a targeted campaign — validated by an A/B holdout.

Modelling approach: rigorous, reproducible, honest

01

Clean & encode

Impute missing values, one-hot encode categoricals, engineer ratios (e.g. minutes-per-equipment-day). 99 model features.

02

Train 4 model families

Logistic Regression (baseline), Random Forest, XGBoost and LightGBM — from interpretable to high-performance.

03

Validate honestly

75/25 stratified split + 5-fold cross-validation on the winner to confirm the score is stable, not luck.

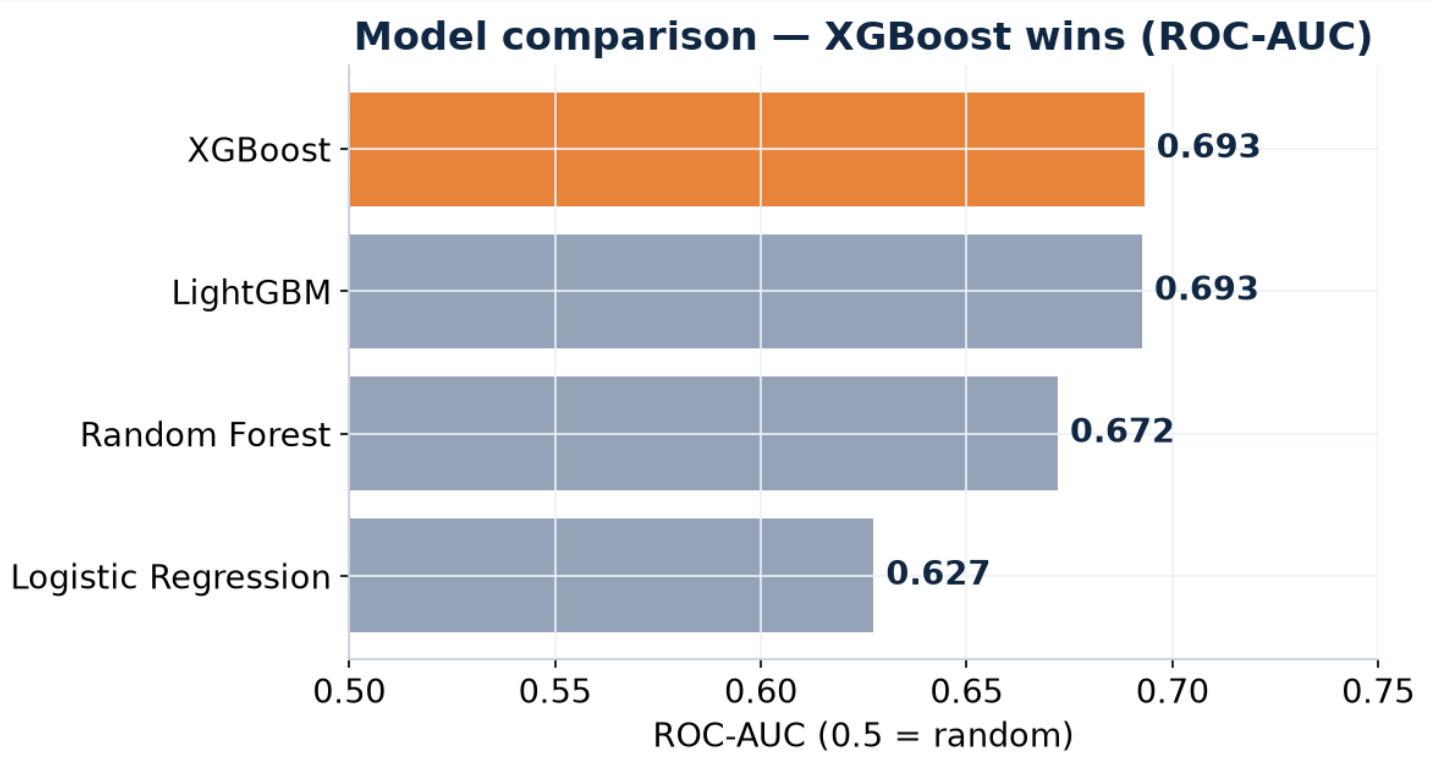
04

Rank by value

Convert probabilities to Expected Value-at-Risk and simulate a targeted retention campaign against random and churn-only baselines.

Cross-validation result: XGBoost ROC-AUC = 0.696 ± 0.004 across 5 folds — confirming the model generalises.

Results: XGBoost wins — AUC 0.693

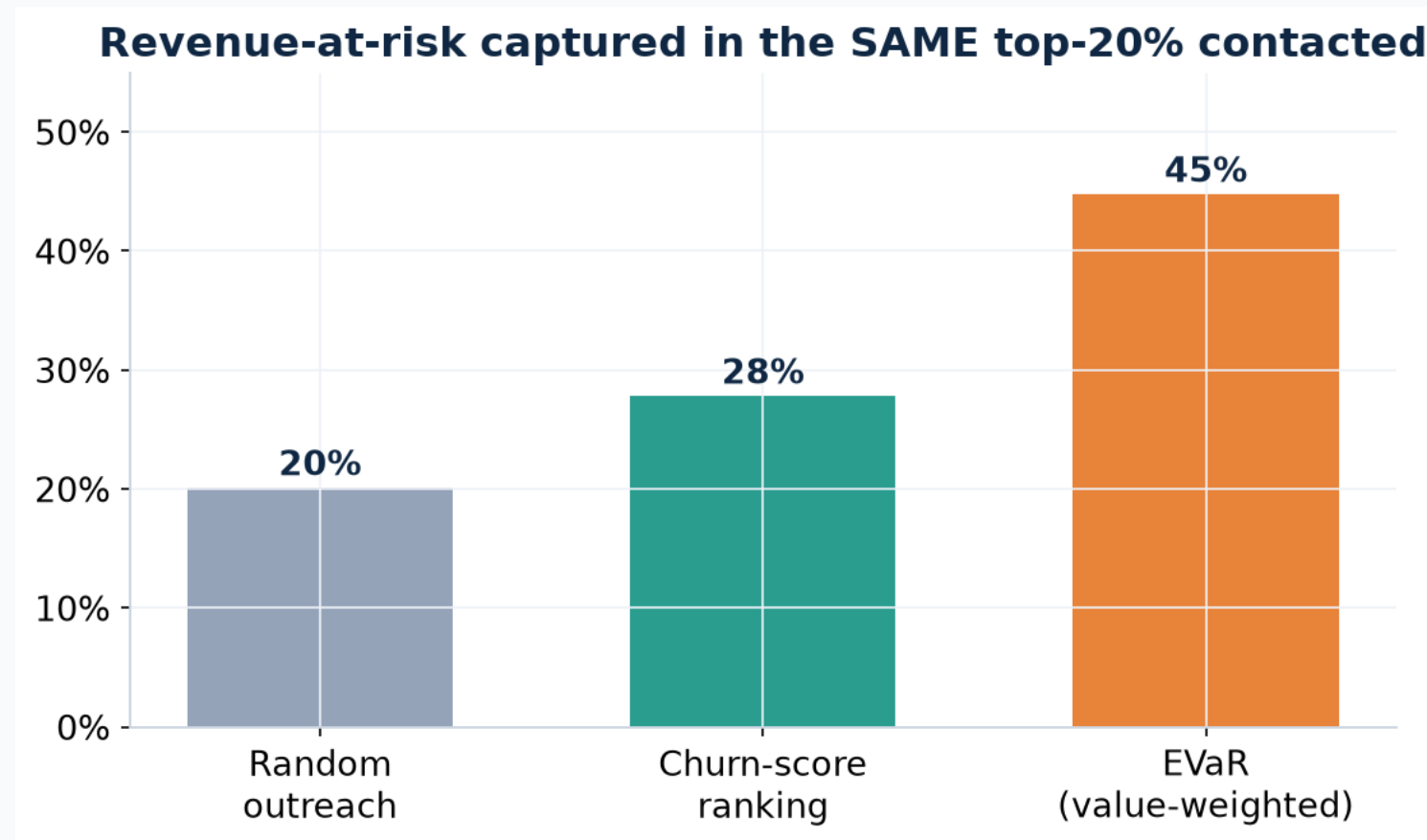


Model	AUC	Acc	Prec	Recall
Logistic Regression	0.627	0.593	0.590	0.586
Random Forest	0.672	0.619	0.609	0.645
LightGBM	0.693	0.635	0.630	0.642
XGBoost	0.693	0.637	0.630	0.648

How to read AUC: 0.5 = guessing, 1.0 = perfect. At 0.693 the model ranks a churner above a non-churner ~69% of the time — enough to target retention spend profitably. Tree models clearly beat the linear baseline.

Source: held-out test set (25% of 100,000), Company A dataset.

Our differentiator: target value, not just probability



The insight

Ranking by churn probability alone wastes budget on low-value churners. Ranking by Expected Value-at-Risk protects the revenue that actually matters.

45%

of revenue-at-risk captured in top 20%

vs 28%

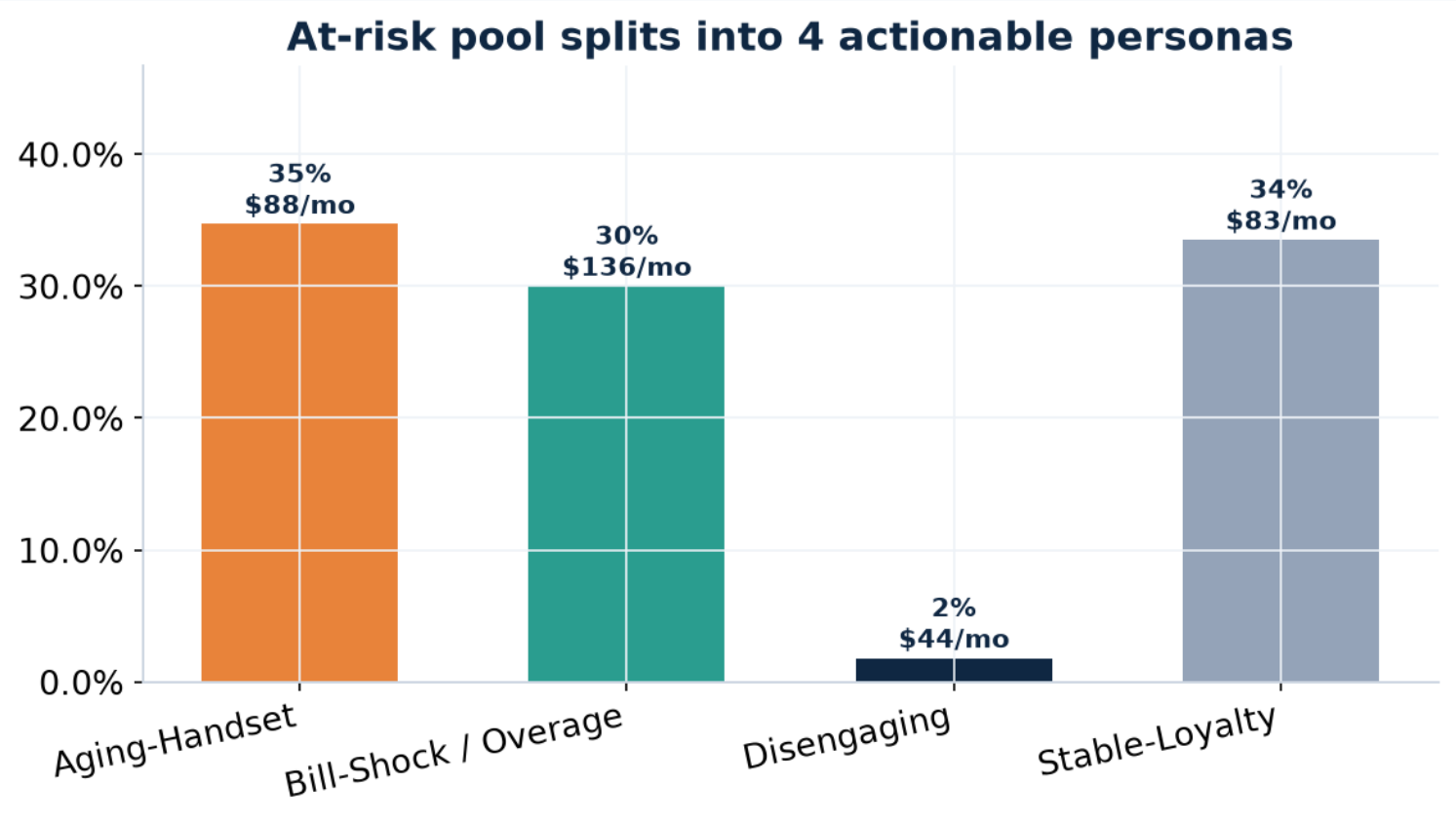
churn-only ranking

1.6x more revenue protected

Same campaign budget, same 20% of customers contacted — but we shield 1.6x the revenue by leading with value.

Source: gains simulation on held-out test set; $EVaR = P(\text{churn}) \times \text{annualised customer revenue}$.

Who to target: four actionable retention personas



Aging-Handset

61% churn · \$88/mo

Old device, decent spend
→ Proactive upgrade / trade-in offer

Bill-Shock / Overage

53% churn · \$136/mo

High bill, overage pain
→ Right-size the plan, autopay discount

Disengaging

80% churn · \$44/mo

Usage falling fast
→ Win-back call + loyalty perk

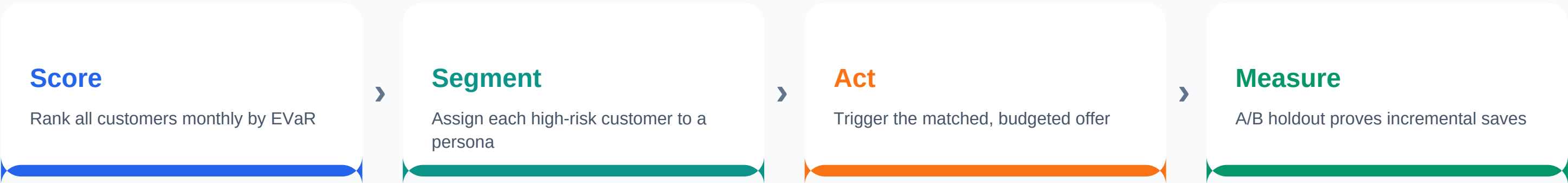
Stable-Loyalty

57% churn · \$83/mo

Healthy, lower risk
→ Light-touch loyalty, protect margin

Source: K-means style segmentation of the high-risk pool, Company A dataset.

The solution: a monthly value-based retention engine



■ **Persona-matched offers**

Aging-handset → upgrade; bill-shock → plan right-sizing; disengaging → win-back. The model decides who; the persona decides what.

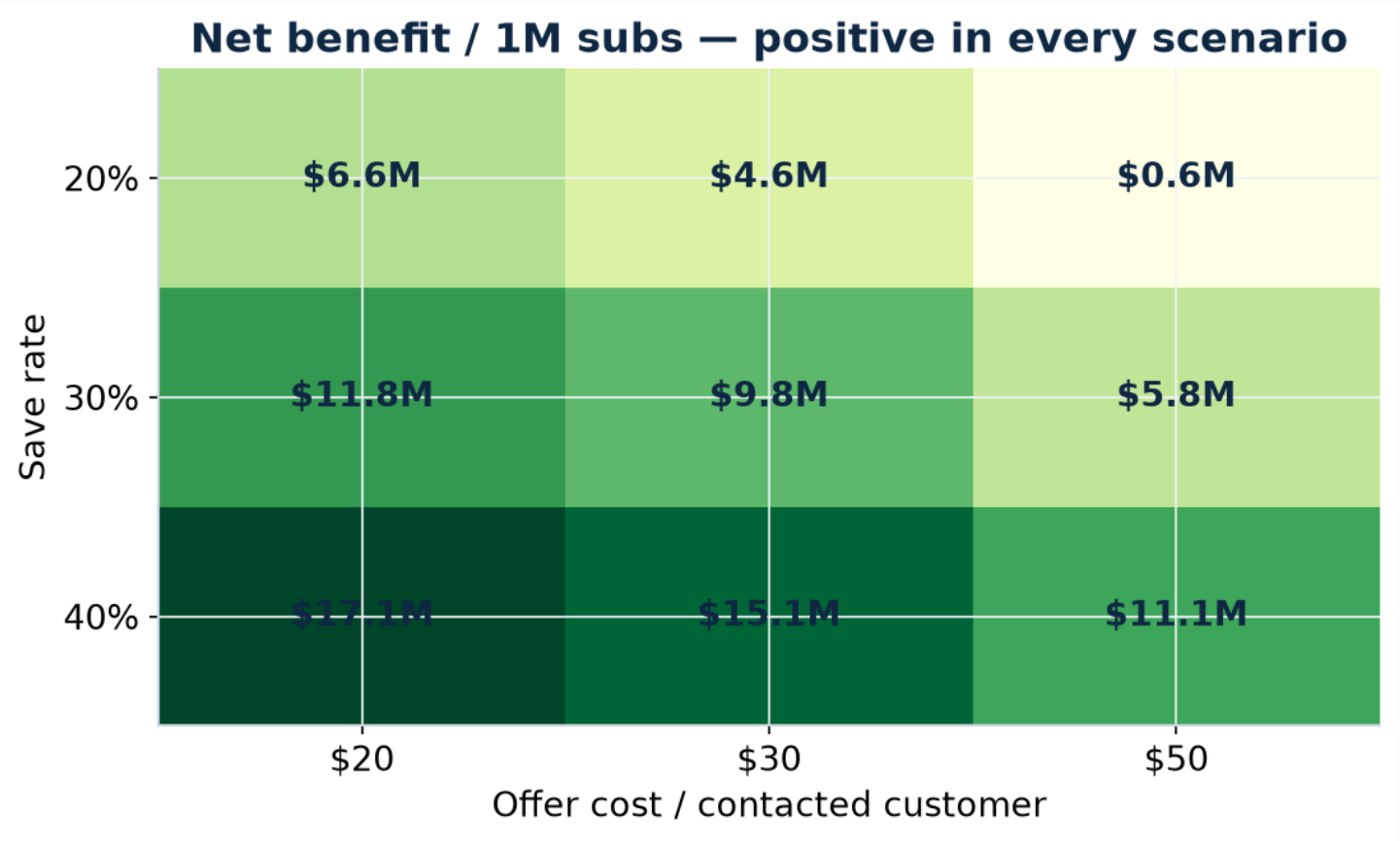
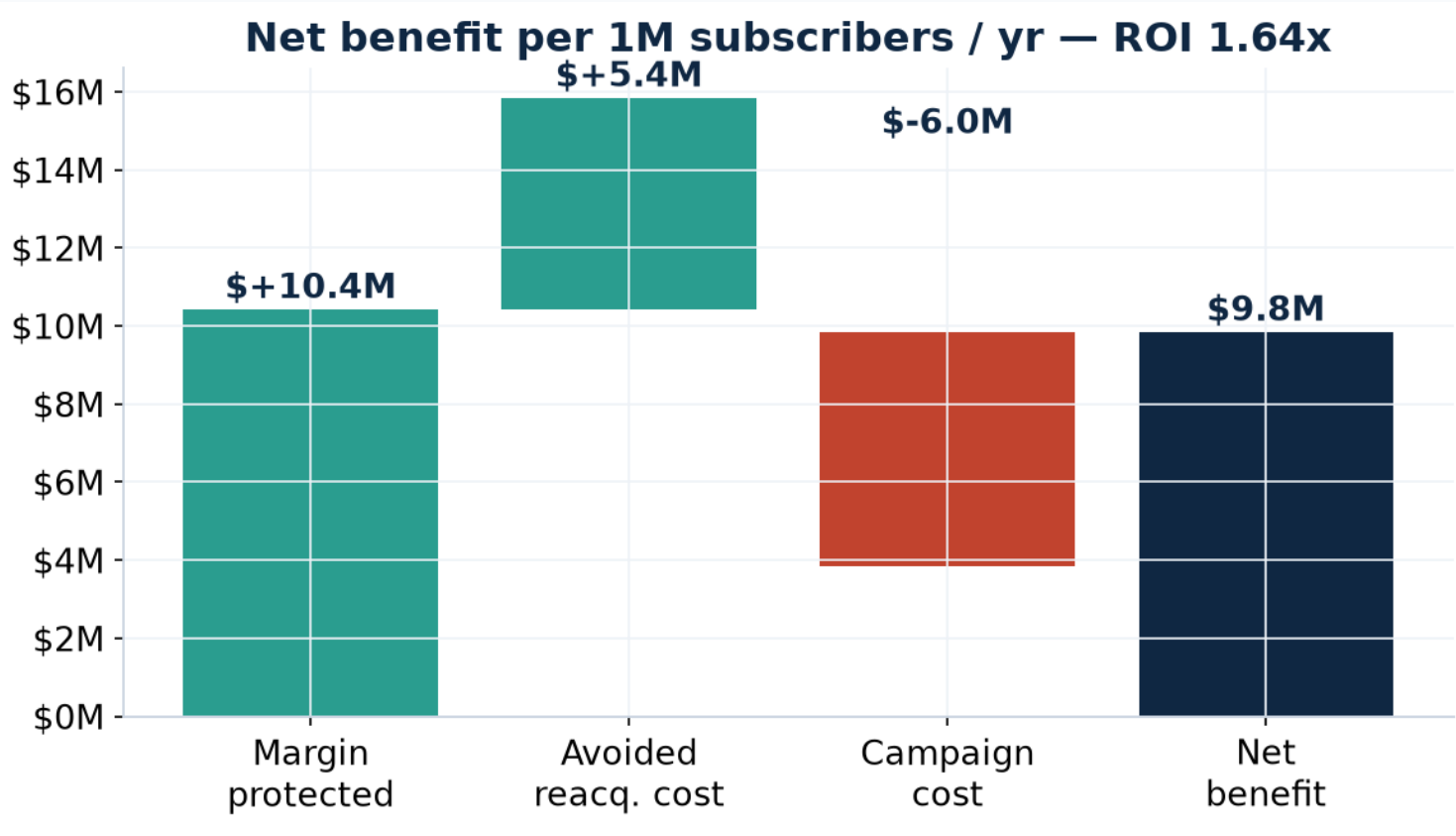
■ **Always-on, not one-off**

Re-scored every month and fed by the A/B holdout, the engine keeps learning which offers actually move retention.

■ **Budget guardrails**

Only the top 20% by EVaR get an offer, so spend is concentrated where it earns a return — never sprayed across the base.

The business case: ROI-positive and robust



\$9.8M
Net benefit / yr

1.64x
Return on spend

15,490
Customers saved / yr

every case
Stays positive

Assumes 22% annual churn, 50% margin, 30% save-rate, \$30 blended offer, \$350 CAC. Sensitivity grid (right) is net of all costs across save-rate x offer-cost.

Our proposal & quotation

We propose a 12-week build of the retention engine, then an ongoing managed service. Our fee is a small fraction of the value we protect.

Phase 1 · Build (wk 1–6)

Data pipeline, model, EVaR scoring, persona logic

Phase 2 · Pilot (wk 7–12)

A/B holdout on top-20% pool; validate incremental saves

Phase 3 · Scale (ongoing)

Monthly scoring, offer optimisation, quarterly review

Investment

One-time setup & build **\$300k**

Managed service / month **\$45k**

Total year 1 **\$840k**

= just 8.5% of the \$9.8M net value we create

Company A keeps ~\$9.0M / yr per 1M subscribers

Fee illustrative for a PoC engagement; scales with subscriber base. Value figures per 1M subscribers / yr.

Risks, fairness & references

■ **Validate before scaling**

The 50/50 sample inflates apparent churn; the A/B pilot calibrates real save-rates and ROI before any full roll-out.

■ **Fairness & compliance**

Demographic fields (ethnicity, income) are de-emphasised to avoid discriminatory targeting; offers are driven by behaviour Company A controls.

■ **Monitor for drift**

Re-train on fresh data and watch feature drift; a stale model silently loses accuracy as customer behaviour shifts.

■ **Don't over-discount**

Budget guardrails and holdouts prevent giving margin away to customers who would have stayed anyway.

- Reichheld & Sasser, “Zero Defections,” Harvard Business Review.
- Bain & Company — customer retention & profitability research.
- Verizon / T-Mobile investor reports, 2024–2025 (churn & ARPU).
- CTIA Annual Wireless Industry Survey, 2025 (market size).
- Neslin et al., “Churn Modeling,” Journal of Marketing Research.
- scikit-learn; XGBoost; LightGBM documentation.

AI-use disclosure: generative AI assisted with code scaffolding, drafting and visualisation. All analysis, numbers and conclusions were verified against the dataset by the author.

Thank you. We look forward to helping Company A win back its base.